Xinran Zhao

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RESEARCH FOCUS

I build language model agents that seek information, plan over long horizons, and work with people when autonomous execution is not enough. My research spans retrieval and deep research, planning and coding agents, and persuasion and human-AI alignment. I study how agents adapt to users' goals, combine external evidence with their own reasoning, and recognize when to ask for help.

EDUCATION

2023– **Ph.D. candidate, Language Technologies Institute (LTI)**
Carnegie Mellon University, PA
Advisor: Tongshuang Wu

2021–23 **M.S. in Computer Science**
Stanford University, CA
Advisor: Christopher Manning; *GPA*: 3.95

2016–20 **B.Eng. in Computer Science and Engineering**
Hong Kong University of Science and Technology, Hong Kong
Advisors: Yangqiu Song and Dit-Yan Yeung; *GPA*: 3.96

2018–19 **Exchange student in Computer Science and Engineering**
Cornell University, NY
Advisor: Claire Cardie

PROFESSIONAL EXPERIENCE

2026,24 **Google DeepMind**, Student Researcher [M.5, P.5, PT.1]
with Azade Nova and Hanie Sedghi;
Research on LLM planning, long-horizon code generation, and continual learning.
May - Nov 2024; May - Nov 2026, Mountain View

2025 **The Allen Institute for Artificial Intelligence**, Research Intern [P.2]
with Varsha Kishore, Aakanksha Naik, Joseph Chang, Jay DeYoung, and Jena Hwang;
Research on deep research agents and attributed long-form report generation.
May 2025 - Dec 2025, Seattle

2023,22 **Tencent AI Lab @ Bellevue**, Research Intern in NLP [P.15, P.9]
with Hongming Zhang, Xiaoman Pan, Wenlin Yao, Dong Yu, and Jianshu Chen;
Research on retrieval-augmented generation, honesty, and hallucination in LLMs.
June 2022 - Sep 2022; June 2023 - Aug 2023, Bellevue

2021-23 **Stanford NLP Group**, Research Assistant [P.19, W.3]
with Dr. Shikhar Murty and Prof. Christopher Manning
Oct 2021 - June 2023, Stanford

2020-21 **HKUST ML Group**, Research Assistant [P.17, W.6]
with Prof. Dit-Yan Yeung
Sep 2020 - Feb 2021, HKUST

2019-21 **HKUST KnowComp Group**, Research Assistant [P.25, P.23, P.21, P.20]
with Prof. Yangqiu Song
June 2019 - August 2021, HKUST

2019-20 **Cornell NLP Group**, Research Assistant [P.22]
with Dr. Esin Durmus and Prof. Claire Cardie
Oct 2019 - August 2020, Cornell

PUBLICATIONS

Manuscripts and Preprints

* denotes equal contribution

2026 M.1 Binjie Wang*, **Xinran Zhao***, Shoubin Yu, Xiang Li, Silong Yong, Chenyang Yang, Tianyu Chen, Ruofei Du, He He, Chen Zhao, and Tongshuang Wu. 3DCodeEdit: Benchmarking Instruction-Guided 3D Program Editing. In submission, 2026

M.2 Danqing Wang, Yuante Li, **Xinran Zhao**, Fei Fang, and Lei Li. Planning while Listening and Speaking for Real-Time Competitive Debate. In submission, 2026

M.3 Yilin Zhang, **Xinran Zhao**, Tong Chen, Ziming Yuan, and Tongshuang Wu. How Should Planning Agents ReSHAPE Their Search Space?. In submission; earlier version presented at BiAlign Workshop at NeurIPS 2026 (Oral), 2026

M.4 **Xinran Zhao**, Xiang Li, Tong Chen, Jinchuan Tian, Zichun Yu, and Tongshuang Wu. Reading the Tea Leaves, Again: Can Large Language Models Predict Language Model Training Runs?. In submission, 2026

M.5 **Xinran Zhao**, Hanie Sedghi, Azade Nova, and Tongshuang Wu. Pivot-ICL: Adaptive Exemplar Selection for In-Context Learning. In submission, 2026

M.6 Tianyu Chen*, **Xinran Zhao***, Vijay Viswanathan, Shoubin Yu, Mingyuan Zhou, and Tongshuang Wu. Knowing When to Stop: The Evidence–Action Gap in Coding Agents. In submission, 2026

M.7 Chenyang Yang, **Xinran Zhao**, Tongshuang Wu, and Christian Kästner. Better Harnesses, Smaller Models: Building 90% Cheaper Agents via Automated Harness Adaptation. In submission, 2026

M.8 Tianyu Chen, Yasi Zhang, Ruiyi Wang, **Xinran Zhao**, Taoran Li, and Mingyuan Zhou. Adaptive-GEPA: Make Your Harness Fit Heterogeneous Requests. In submission, 2026

2025 M.9 **Xinran Zhao**, Boyuan Zheng, Chenglei Si, Haofei Yu, Ken Liu, Runlong Zhou, Ruochen Li, Tong Chen, Xiang Li, Yiming Zhang, and Tongshuang Wu. The Ramon Llull's Thinking Machine for Automated Ideation. In submission; presented at LM4Sci at COLM 2025, 2025

M.10 Yixiao Zeng, Tianyu Cao, Danqing Wang, **Xinran Zhao**, Zimeng Qiu, Morteza Ziyadi, Tongshuang Wu, and Lei Li. RARE: Retrieval-Aware Robustness Evaluation for Retrieval-Augmented Generation Systems. In submission, 2025

Peer-reviewed Conference and Journal Publications

* denotes equal contribution

2026 P.1 Rulin Shao, Akari Asai, Shannon Zejiang Shen, Hamish Ivison, Varsha Kishore, Jingming Zhuo, **Xinran Zhao**, Molly Park, Samuel G. Finlayson, David Sontag, Tyler Murray, Sewon Min, Pradeep Dasigi, Luca Soldaini, Faeze Brahman, Wen tau Yih, Tongshuang Wu, Luke Zettlemoyer, Yoon Kim, Hannaneh Hajishirzi, and Pang Wei Koh. DR Tulu: Reinforcement Learning with Evolving Rubrics for Deep Research. In *Proceedings of the 43rd International Conference on Machine Learning*,
ICML 2026 (Spotlight)

P.2 **Xinran Zhao**, Aakanksha Naik, Jay DeYoung, Joseph Chee Chang, Jena D. Hwang, Tongshuang Wu, and Varsha Kishore. Improving Attributed Long-form Question Answering with Intent Awareness. In *Proceedings of the 14th International Conference on Learning Representations*,
ICLR 2026

P.3 Danqing Wang*, Zhuorui Ye*, **Xinran Zhao***, Fei Fang, and Lei Li. Strategic Planning and Rationalizing on Trees Make LLMs Better Debaters. In *Proceedings of the 14th International Conference on Learning Representations*,
ICLR 2026

P.4 Fengyu Cai, Tong Chen, **Xinran Zhao**, Sihao Chen, Hongming Zhang, Tongshuang Wu, Iryna Gurevych, and Heinz Koeppl. Revela: Dense Retriever Learning via Language Modeling. In *Proceedings of the 14th International Conference on Learning Representations*,
ICLR 2026 (Oral). Best Paper Award, FrontierIR Workshop at AAAI 2026

2025 P.5 **Xinran Zhao**, Hanie Sedghi, Bernd Bohnet, Dale Schuurmans, and Azade Nova. Improving Large Language Model Planning with Action Sequence Similarity. In *Proceedings of the 13th International Conference on Learning Representations*,
ICLR 2025

P.6 Jushaan Singh Kalra*, **Xinran Zhao***, To Eun Kim, Fengyu Cai, Fernando Diaz, and Tongshuang Wu. MoR: Better Handling Diverse Queries with a Mixture of Sparse, Dense, and Human Retrievers. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*,
EMNLP 2025

P.7 Yilin Zhang, **Xinran Zhao**, Zora Zhiruo Wang, Chenyang Yang, Jiayi Wei, and Tongshuang Wu. cAST: Enhancing Code Retrieval-Augmented Generation with Structural Chunking via Abstract Syntax Tree. In *Findings of the 2025 Conference on Empirical Methods in Natural Language Processing*,
EMNLP 2025

P.8 Qianou Ma*, Dora Zhao*, **Xinran Zhao**, Chenglei Si, Chenyang Yang, Ryan Louie, Ehud Reiter, Diyi Yang+, and Tongshuang Wu+. SPHERE: An Evaluation Card for Human-AI Systems. In *Findings of the 63rd Annual Meeting of the Association for Computational Linguistics*, ACL 2025

2024 P.9 **Xinran Zhao**, Hongming Zhang, Xiaoman Pan, Wenlin Yao, Dong Yu, Tongshuang Wu, and Jianshu Chen. Fact-and-Reflection (FaR) Improves Confidence Calibration of Large Language Models. In *Findings of the 62nd Annual Meeting of the Association for Computational Linguistics*, ACL 2024

P.10 Fengyu Cai, **Xinran Zhao**, Hongming Zhang, Iryna Gurevych, and Heinz Koeppl. GeoHard: Towards Measuring Class-wise Hardness through Modelling Class Semantics. In *Findings of the 62nd Annual Meeting of the Association for Computational Linguistics*, ACL 2024

P.11 **Xinran Zhao**, Tong Chen, Sihao Chen, Hongming Zhang, and Tongshuang Wu. Beyond Relevance: Evaluate and Improve Retrievers on Perspective Awareness. In *Proceedings of the 1st Conference on Language Modeling*, COLM 2024

P.12 Tong Chen, Hongwei Wang, Sihao Chen, Wenhao Yu, Kaixin Ma, **Xinran Zhao**, Hongming Zhang, and Dong Yu. Dense X Retrieval: What Retrieval Granularity Should We Use?. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, EMNLP 2024

P.13 Cheng Qian, **Xinran Zhao**, and Tongshuang Wu. "Merge Conflicts!" Exploring the Impacts of External Distractors to Parametric Knowledge Graphs. In *Proceedings of the 1st Conference on Language Modeling*, COLM 2024

P.14 Fengyu Cai, **Xinran Zhao**, Tong Chen, Sihao Chen, Hongming Zhang, Iryna Gurevych, and Heinz Koeppl. MixGR: Enhancing Retriever Generalization for Scientific Domain through Complementary Granularity. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, EMNLP 2024

2023 P.15 **Xinran Zhao**, Hongming Zhang, Xiaoman Pan, Wenlin Yao, Dong Yu, and Jianshu Chen. Thrust: Adaptively Propels Large Language Models with External Knowledge. In *Proceedings of the 37th Conference on Neural Information Processing Systems*, NeurIPS 2023

P.16 Xiang Li, Jiangwei Yu, **Xinran Zhao**, Hongming Zhang, and Yu-Xiong Wang. Video State-changing Object Segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, ICCV 2023

P.17 **Xinran Zhao**, Esin Durmus, and Dit-Yan Yeung. Towards Reference-free Text Simplification Evaluation. In *Findings of the 61st Annual Meeting of the Association for Computational Linguistics*, ACL 2023

P.18 BIG-bench collaboration (including **Xinran Zhao**). Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models. In *Transactions on Machine Learning Research*, TMLR 2023

2022 P.19 **Xinran Zhao***, Shikhar Murty*, and Christopher Manning. On Measuring the Intrinsic Few-Shot Hardness of Datasets. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*,
EMNLP 2022

P.20 **Xinran Zhao**, Hongming Zhang, and Yangqiu Song. PCR4ALL: A Comprehensive Evaluation Benchmark for Pronoun Coreference Resolution. In *Proceedings of the Language Resources and Evaluation Conference*,
LREC 2022

P.21 Ziqian Zeng, Weimin Ni, Tianqing Fang, Xiang Li, **Xinran Zhao**, and Yangqiu Song. Weakly Supervised Text Classification Using Supervision Signals from a Language Model. In *Findings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*,
NAACL 2022

2021 P.22 **Xinran Zhao**, Esin Durmus, Hongming Zhang, and Claire Cardie. Leveraging Topic Relatedness for Argument Persuasion. In *Findings of the 59th Annual Meeting of the Association for Computational Linguistics*,
ACL 2021

P.23 Nedjma Ousidhoum, **Xinran Zhao**, Tianqing Fang, Yangqiu Song, and Dit-Yan Yeung. Probing Toxic Content in Large Pre-Trained Language Models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*,
ACL 2021

P.24 Yue Xi, Jiale Huo, **Xinran Zhao**, Yushi Jiang, and Qiang Yang. The Effects of Fear of Missing Out on Social Media Posting Preferences. In *European Journal of Marketing*,
EJM 2021

2020 P.25 Hongming Zhang*, **Xinran Zhao***, and Yangqiu Song. WinoWhy: A Deep Diagnosis of Essential Commonsense Knowledge for Answering Winograd Schema Challenge. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*,
ACL 2020

Workshop Papers, Technical Reports, and Extended Abstracts

2026 W.1 Zora Zhiruo Wang, Apurva Gandhi, Rulin Shao, Aspen Chen, Jonas Mueller, Zhiqi Liang, Jett Chen, Michael Ryan, Qianou Ma, Luxi He, Zhoujun Cheng, Andre He, Seungone Kim, Jiayi Geng, Mingqian Zheng, Weiwei Sun, Zheyuan Zhang, **Xinran Zhao**, Yike Wang, Abe Hou, Liwei Jiang, Pang Wei Koh, Diyi Yang, Graham Neubig, and Daniel Fried. Efficient Test-Time Adaptation through Human-AI Interaction. In *Test-Time and Continual Learning Workshop at NeurIPS 2026*,
Poster

2024 W.2 Zirui Wang, **Xinran Zhao**, Simon Stepputtis, Woojun Kim, Tongshuang Wu, Katia P. Sycara, and Yaqi Xie. HiMemFormer: Hierarchical Memory-Aware Transformer for Multi-Agent Action Anticipation. In *VLM Workshop @ 38th Conference on Neural Information Processing Systems*,
VLM @ NeurIPS 2024

2022 W.3 **Xinran Zhao**(lead), Tina Li, Xiaoyuan Ni, and Christopher Manning (advising). Can Pre-trained Language Models Understand Definitions?. In *CS 224N Final Project*,
Stanford 2022

2021 W.4 Hongming Zhang, **Xinran Zhao**, and Yangqiu Song. A Brief Survey and Comparative Study of Recent Development of Pronoun Coreference Resolution in English. In *4th Workshop on Computational Models of Reference, Anaphora and Coreference @ Conference on Empirical Methods in Natural Language Processing*,
CRAC @ EMNLP 2021

W.5 Hongming Zhang, Yintong Huo, **Xinran Zhao**, Yangqiu Song, and Dan Roth. Learning Contextual Causality between Daily Events from Time-consecutive Images. In *Causality in Vision Workshop @ IEEE/CVF Conference on Computer Vision and Pattern Recognition*,
CIV @ CVPR 2021

W.6 **Xinran Zhao**. An Online Learning Platform with Natural Language Processing Supported Teaching Assistant. In *Final Year Project, advised by Dit-Yan Yeung*,
HKUST 2021

Patents

2025 PT.1 **Xinran Zhao**, Hanie Sedghi, Bernd Bohnet, Dale Schuurmans, and Azade Nova. Improving Large Language Model Planning with Action Sequence Similarity.
US Patent (submitted), 2025

SELECTED HONORS AND AWARDS

2026 Best Paper Award, FrontierIR Workshop at AAAI (Revela) [P.4]

2025 Amazon AI PhD Fellowship (2-year full tuition and stipend; $10,000 AWS credits)

2024 Google RSGCP Award on LLM Planning ($30,000)
Google RS Conference Scholarship Travel ($3,000)

2021 HKUST Academic Achievement Medal (Top 1%)

2016-20 Continuous Undergraduate Scholarship (Top 5%)
First Class Honors, Dean's List for active semesters

MENTORING EXPERIENCE

2024-25 **Yilin Zhang** (CMU MS, now MLE at Google)
Retrieval augmented code generation [P.7].

Jushaan Kalra (CMU MS, now MLE at Snowflake)
Multi-domain retrieval augmented generation [P.6].

Zhuorui Ye (visited CMU, now PhD student at Princeton University)
Agents for Competitive Debates [P.3].

Jack Zeng (CMU MS, now MLE at XPeng)
Scaled time-sensitive multi-domain retrieval evaluation [M.10].

2023 **Cheng Qian** (visited CMU, now PhD student at UIUC)
Knowledge Conflicts between external and parametric model knowledge [P.13].

INVITED TALKS

2026 **Continual Learning for Coding Agents**
Google DeepMind Internal Talk (Summer 2026)

2025 **Improving Large Language Model Planning with Action Sequence Similarity**
H2 Group at UW (Summer 2025)

2024 **Improving Large Language Model Planning with Action Sequence Similarity**
Google DeepMind Internal Talk (Summer 2024)
Towards Building Retrieval Systems in Complex Realistic Scenarios
Samaya AI (Summer 2024)

2022 **On Task Difficulty of Few-shot Learning**
ICML 2022 Commonsense Tutorial (Summer 2022)

NLP as a Tool for Scientific Discovery
Department of Marketing @ the National University of Singapore (Spring 2022)

PROFESSIONAL SERVICE

Teaching Assistant

2026 CMU LTI 11766: Large Language Model Applications

2025 CMU LTI 11763: Inference Algorithms for Language Modeling

2020 HKUST COMP 4332: Big Data Mining and Management (affiliated helper)

Area Chair

NLP **ACL** 2024–current, **EMNLP** 2024–current, **NAACL** 2025–current

Paper Reviewer

NLP **ACL** 2020–23, **EMNLP** 2021–23, **EACL** 2023–24, **NAACL** 2021–24, **AACL** 2023

ML **NeurIPS** 2024–25, **ICLR** 2025, **AAAI** 2023–24, **AKBC** 2021

CV **ECCV** 2022

Updated October 2026